

ENERGY METER

TEAM 12

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OBJECTIVES

- To design a synthesizable digital energy meter using Verilog HDL.
- To implement a Finite State Machine (FSM)-based architecture for energy calculation.
- To verify the functionality using Cadence simulation tools.
- To implement and understand the complete ASIC design flow.
- To analyze energy computation using digital accumulation techniques.

Digital Energy Meter

RTL Flow — 3 Step Operation

Inputs

Voltage [3:0] - Current [3:0]

Step 1 — Power Calculation

Power = Voltage $\times$ Current
Stored in 8-bit register

Step 2 — Energy Calculation

Energy = Power $\times$ Time (10 ns/clock)
Stored in 12-bit register

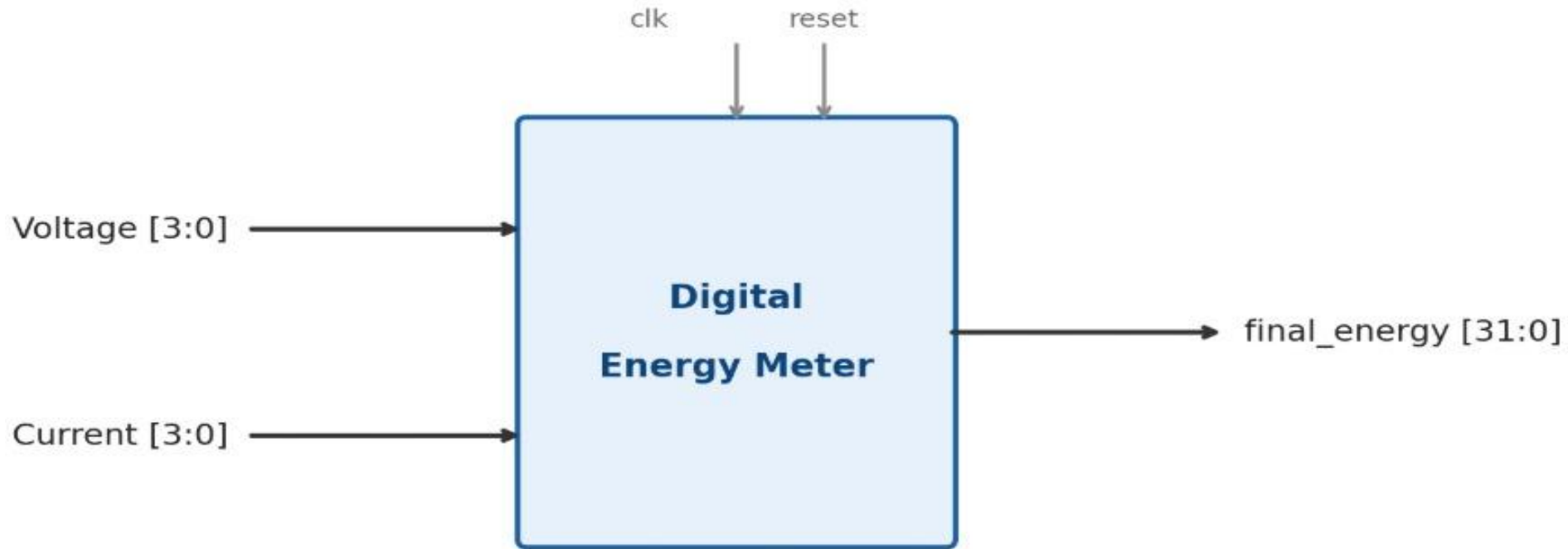
FSM
cycles

Step 3 — Energy Accumulation

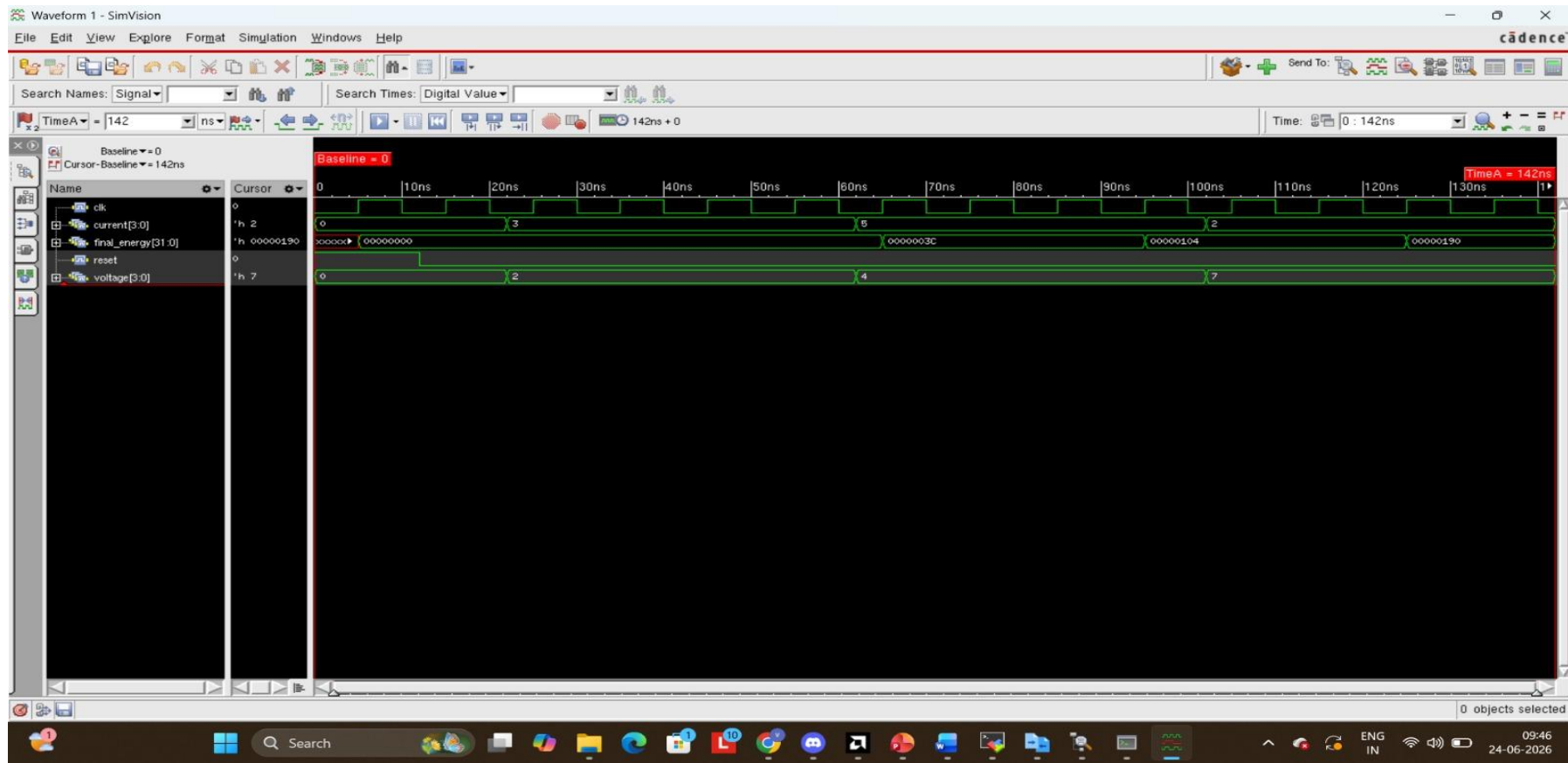
final_energy += energy
Running total in 32-bit register

Output: final_energy [31:0]

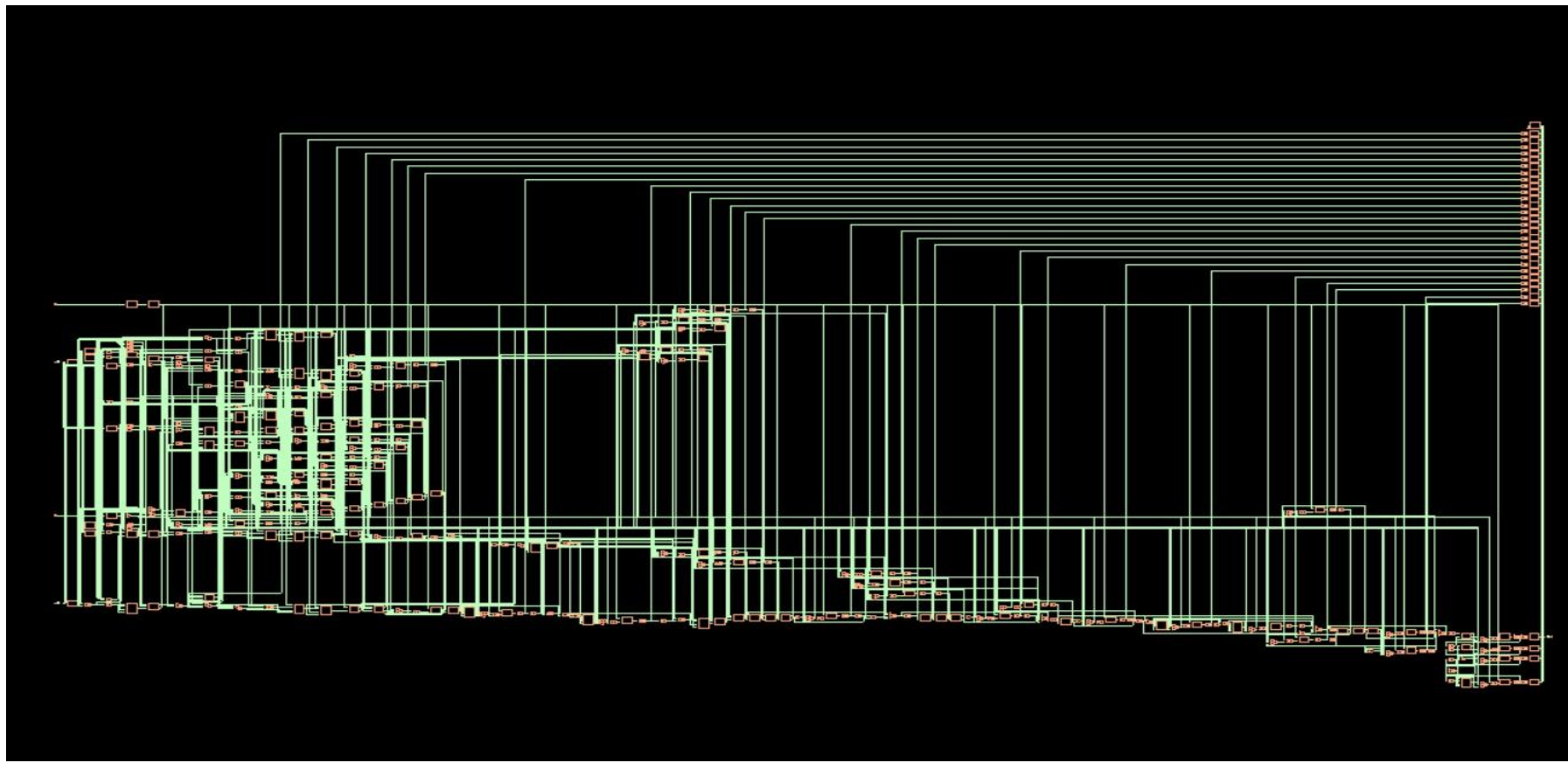
BLOCK DIAGRAM



SIMULATION



SYNTHESIS

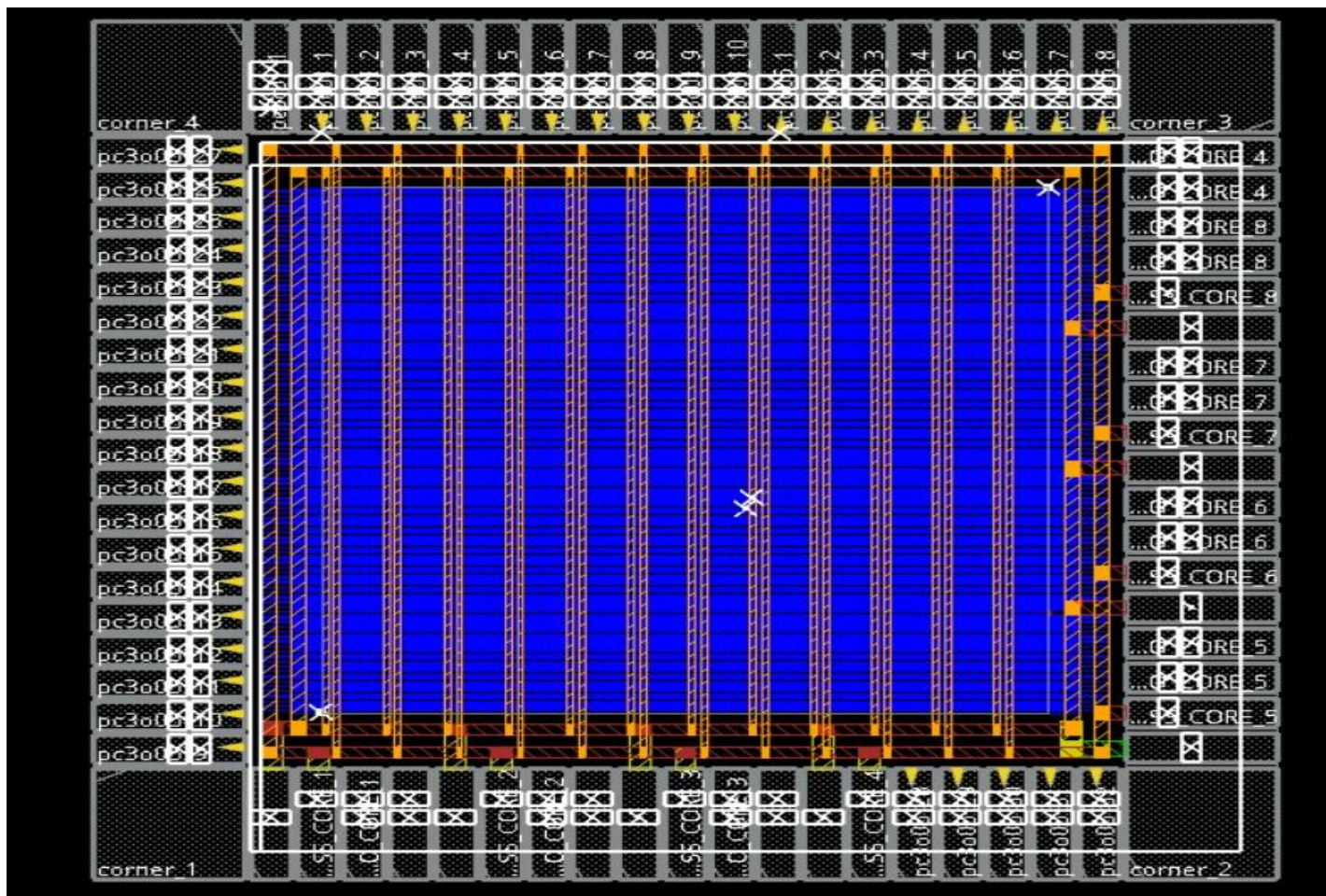


Metric	Value
Total Area (μm^2)	67,105.694
Total Power (mW)	10.0566
Total Slack (ns)	-1.974

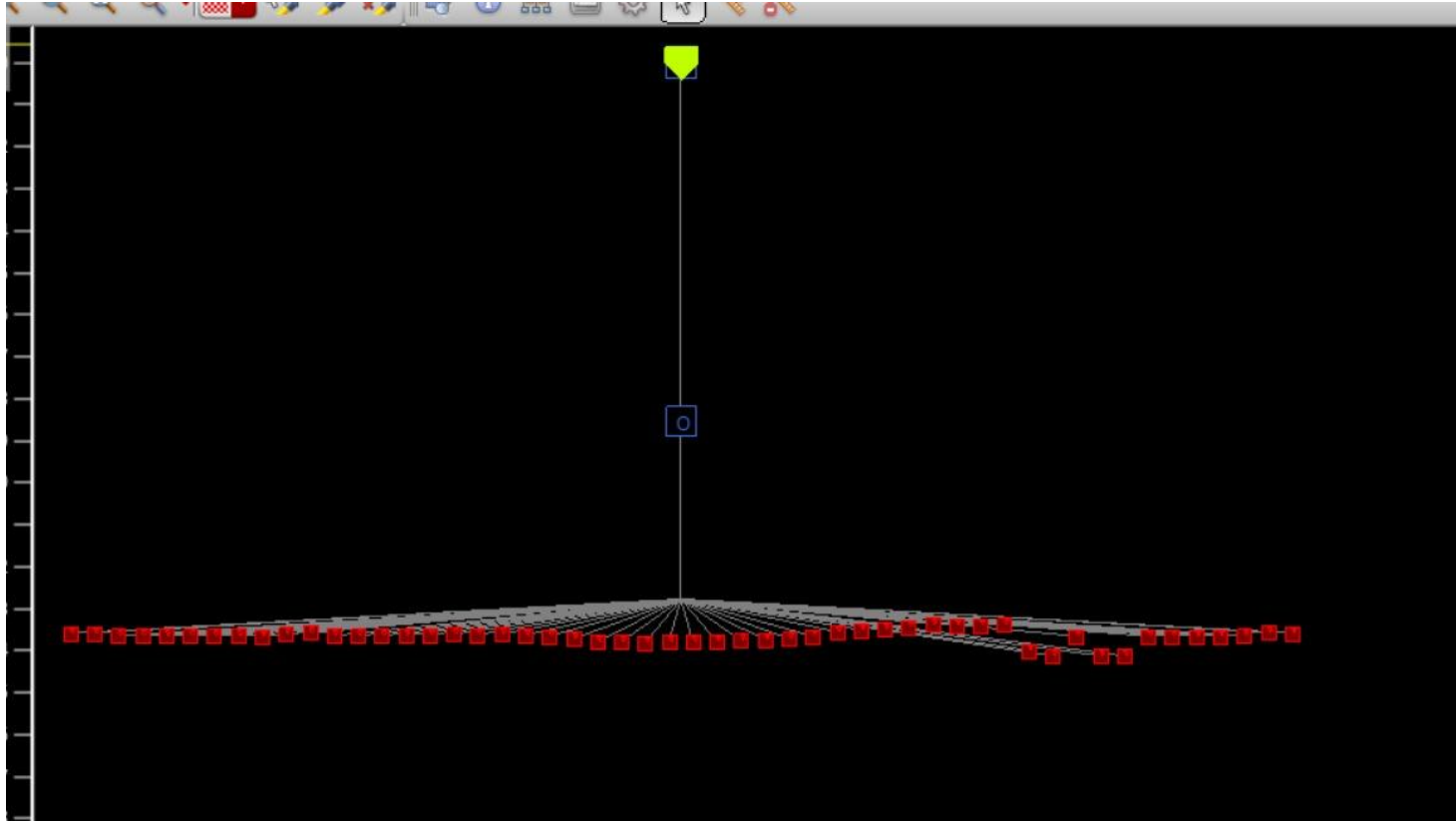
FLOORPLAN



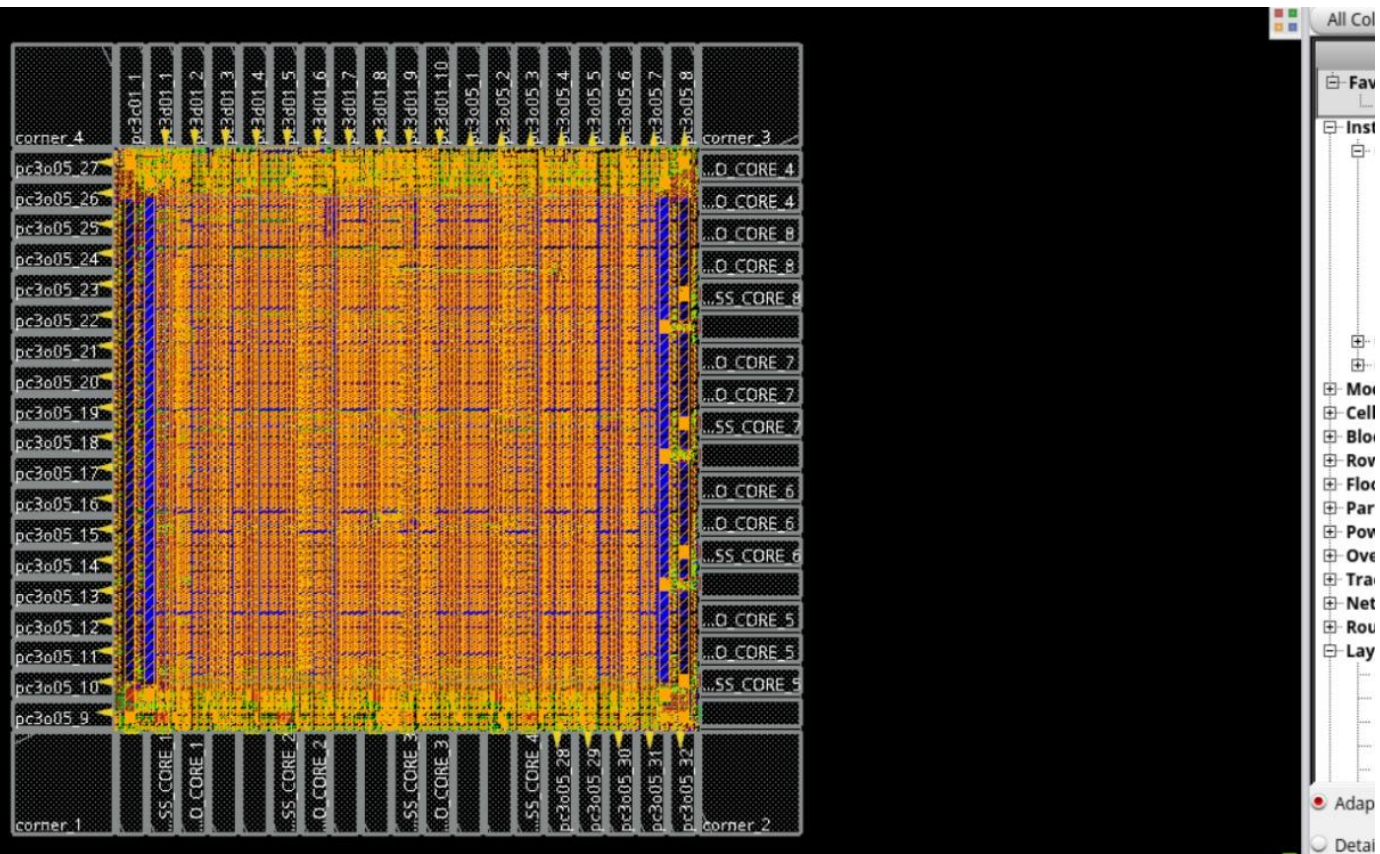
POWERPLAN



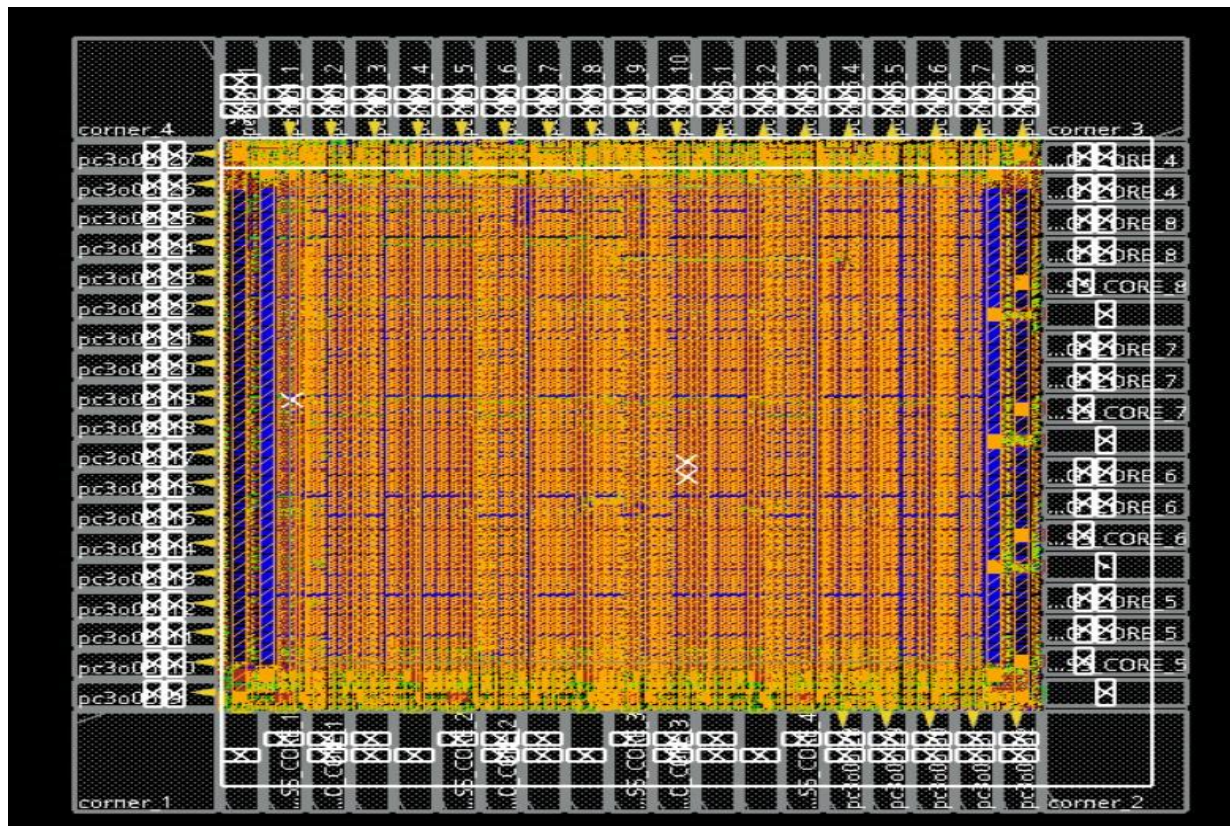
CLOCK TREE SYNTHESIS



ROUTING



SIGNOFF





THANK YOU

